

Latent Geometry of Synthetic Neural Manifolds

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ABSTRACT

We study how latent representations fold under cohort-level validation across synthetic neural manifolds. Recent work suggests manifold curvature predicts generalization better than parameter count [1,2]. We report three findings across twelve fictional benchmarks and release all code under a permissive license.

1. INTRODUCTION

Scientific peer review should focus on claims, not formatting. Our baseline comparison against prior manifold-learning methods remains limited to synthetic tasks. Topological summaries align with Chen et al. [1] but require clearer notation for readers unfamiliar with persistent homology.

2. METHODS

We estimate manifold curvature from Jacobian spectra on held-out folds. Each cohort-level validation split uses disjoint subject IDs. Hyperparameters follow Vale & Mensch [2] unless noted otherwise.

References

1. Chen, R., Hale, T. Geometry of latent folds. *Journal of Fictional ML* 4(1), 2024.
<https://doi.org/10.5555/fictional-chen-2024>
2. Vale, K., Mensch, K. Curvature and generalization in synthetic manifolds. *Proc. Fictional AI*, 2023.
<https://doi.org/10.5555/fictional-vale-2023>